

Amazon Networking Services

Overview

This document introduces **Amazon cloud networking**, which enables communication between cloud resources and with external networks. Networking is foundational to designing secure, scalable AWS architectures.

Topics covered include:

1. **Computer Network Basics** – Core networking concepts such as IP addresses, subnets, and CIDR notation.
2. **Amazon Virtual Private Cloud (VPC)** – The virtual network where AWS resources (for example, EC2) communicate.
3. **AWS Network Security** – Firewalls and access controls used to protect cloud resources.
4. **AWS Direct Connect** – Dedicated private connectivity between on-premises data centers and AWS.
5. **Amazon DNS (Route 53)** – Scalable domain name resolution and traffic routing.
6. **Amazon CDN (CloudFront)** – Low-latency global content delivery.

Understanding these services builds a strong foundation for designing and operating secure AWS cloud environments.

Reviewing Computer Network Basics

A **computer network** consists of interconnected devices that share resources and exchange data. A **subnet** is a logical subdivision of a network, created to improve performance, security, and manageability.

Each subnet is identified by a **network address**, and every device within it is assigned a unique **IP address** to enable communication.

IP Address

An **IP (Internet Protocol) address** uniquely identifies a device on a network and enables data exchange. Two versions exist:

- **IPv4** – 32-bit addresses written as four decimal numbers (for example, **172.217.9.36**)
- **IPv6** – 128-bit hexadecimal addresses

This chapter focuses on IPv4.

Figure 3.1 illustrates a sample website and its IPv4 address.

zeebestbuy.com

172.217.9.36

10101100. 11011001. 00001001.00100100

Figure 3.1 – An IP address

An IPv4 address contains **32 bits (4 bytes)**, allowing a theoretical total of 2^{32} addresses. Since a single network cannot practically contain all these addresses, networks are divided into subnets using **CIDR**.

CIDR

Classless Inter-Domain Routing (CIDR) defines IP address ranges and subnet sizes. CIDR notation consists of an IP address followed by a slash (/) and the number of fixed network bits.

Example: 172.217.9.0/24

- First 24 bits: network portion
- Remaining 8 bits: host portion

This subnet contains **256 IP addresses** (172.217.9.0 to 172.217.9.255).

Figure 3.2 shows common CIDR ranges and address capacities.

Subnet Mask Hierarchy

Subnet Mask	CIDR	Binary Notation	Available Addresses Per Subnet
255.255.255.255	/32	11111111.11111111.11111111.11111111	1
255.255.255.254	/31	11111111.11111111.11111111.11111110	2
255.255.255.252	/30	11111111.11111111.11111111.11111100	4
255.255.255.248	/29	11111111.11111111.11111111.11111000	8
255.255.255.240	/28	11111111.11111111.11111111.11110000	16
255.255.255.224	/27	11111111.11111111.11111111.11100000	32
255.255.255.192	/26	11111111.11111111.11111111.11000000	64
255.255.255.128	/25	11111111.11111111.11111111.10000000	128
255.255.255.0	/24	11111111.11111111.11111111.00000000	256
255.255.254.0	/23	11111111.11111111.11111110.00000000	512
255.255.252.0	/22	11111111.11111111.11111100.00000000	1024
255.255.248.0	/21	11111111.11111111.11111000.00000000	2048
255.255.240.0	/20	11111111.11111111.11110000.00000000	4096
255.255.224.0	/19	11111111.11111111.11100000.00000000	8192
255.255.192.0	/18	11111111.11111111.11000000.00000000	16384
255.255.128.0	/17	11111111.11111111.10000000.00000000	32768
255.255.0.0	/16	11111111.11111111.00000000.00000000	65536
255.254.0.0	/15	11111111.11111110.00000000.00000000	131072
255.252.0.0	/14	11111111.11111100.00000000.00000000	262144
255.248.0.0	/13	11111111.11111000.00000000.00000000	524288
255.240.0.0	/12	11111111.11110000.00000000.00000000	1048576
255.224.0.0	/11	11111111.11100000.00000000.00000000	2097152
255.192.0.0	/10	11111111.11000000.00000000.00000000	4194304
255.128.0.0	/9	11111111.10000000.00000000.00000000	8388608
255.0.0.0	/8	11111111.00000000.00000000.00000000	16777216

Figure 3.2 – CIDR notation

CIDR enables flexible network design and efficient IP address allocation.

The Internet

The **internet** is a global system of interconnected networks that exchange data using standardized protocols such as TCP/IP.

Key components include:

- **Routers** – Forward traffic between networks using route tables
- **Switches** – Connect devices within a network using MAC addresses

Each subnet is associated with exactly **one route table**, which determines how traffic is forwarded.

Modern networks, like compute systems, have evolved from physical hardware to **software-defined virtual networks**, such as Amazon VPC.

Understanding Amazon Virtual Private Cloud

An **Amazon VPC** is a logically isolated virtual network in the AWS cloud. It allows full control over IP addressing, subnets, routing, and security.

Key benefits include:

- 1. **High Availability** – Regional scope with subnets spanning Availability Zones
- 2. **Flexibility** – Custom CIDR ranges, routing, and network topology
- 3. **Isolation** – Separation of workloads (for example, production vs development)
- 4. **Cost Efficiency** – No charge for creating or using VPCs
- 5. **Security** – Subnet-level and instance-level firewalls
- 6. **Hybrid Connectivity** – Integration with on-premises networks via VPN or Direct Connect

Part One – Creating a VPC with Subnets

This section demonstrates VPC and subnet creation using both the AWS Management Console and AWS CloudShell.

Creating a VPC and a Subnet Using the AWS Console

Create a VPC named `my-vpc1` with CIDR `10.10.0.0/16` and no IPv6 block.

VPC settings

Resources to create Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

IPv6 CIDR block Info

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy Info

Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

X

Value - optional

X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create VPC

Figure 3.3 – VPC settings

Create a subnet named `public-subnet` with CIDR `10.10.1.0/24`.

VPC ID

Create subnets in this VPC.

vpc-0194dc489edcfb09b (my VPC1)

Associated VPC CIDRs

IPv4 CIDRs

10.10.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

public-subnet

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1a

IPv4 CIDR block

Info

10.10.1.0/24

Tags - optional

Key

Value - optional

Q Name X

Q public-subnet X

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

Figure 3.4 – Subnet settings

Creating an AWS Cloud Network Using CloudShell

AWS CloudShell provides CLI-based resource management.

1. Create VPC1

```
aws ec2 create-vpc --cidr-block 10.10.0.0/16 \  
--tag-specifications ResourceType=vpc,Tags='[{Key=Name,Value="VPC1"}]'
```

5 / 22

```
[cloudshell-user@ip-10-2-126-119 ~]$ aws ec2 create-vpc \
> --cidr-block 10.10.0.0/16 \
> --tag-specifications ResourceType=vpc,Tags='[{Key=Name,Value="VPC1"}]'
{
  "Vpc": {
    "CidrBlock": "10.10.0.0/16",
    "DhcpOptionsId": "dopt-f8579e9d",
    "State": "pending",
    "VpcId": "vpc-0e6043d6091faa7cd",
    "OwnerId": "317332158300",
    "InstanceTenancy": "default",
    "Ipv6CidrBlockAssociationSet": [],
    "CidrBlockAssociationSet": [
      {
        "AssociationId": "vpc-cidr-assoc-04116e8e5e98b256f",
        "CidrBlock": "10.10.0.0/16",
        "CidrBlockState": {
          "State": "associated"
        }
      }
    ],
    "IsDefault": false,
    "Tags": [
      {
        "Key": "Name",
        "Value": "VPC1"
      }
    ]
  }
}
```

Figure 3.5 – Create VPC1

2. Create Public and Private Subnets

```
aws ec2 create-subnet --vpc-id vpc-xxxx \
--cidr-block 10.10.1.0/24 \
--tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="public-subnet"}]'
```

```
aws ec2 create-subnet --vpc-id vpc-xxxx \
--cidr-block 10.10.2.0/24 \
--tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="private-
subnet"}]'
```

```
[cloudshell-user@ip-10-2-126-119 ~]$ aws ec2 create-subnet \
> --vpc-id vpc-0e6043d6091faa7cd \
> --cidr-block 10.10.2.0/24 \
> --tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="private-subnet"}]'
{
  "Subnet": {
    "AvailabilityZone": "us-west-2a",
    "AvailabilityZoneId": "usw2-az1",
    "AvailableIpAddressCount": 251,
    "CidrBlock": "10.10.2.0/24",
    "DefaultForAz": false,
    "MapPublicIpOnLaunch": false,
    "State": "available",
    "SubnetId": "subnet-0328a7940791d9fc0",
    "VpcId": "vpc-0e6043d6091faa7cd",
    "OwnerId": "317332158300",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "Tags": [
      {
        "Key": "Name",
        "Value": "private-subnet"
      }
    ],
    "SubnetArn": "arn:aws:ec2:us-west-2:317332158300:subnet/subnet-0328a7940791d9fc0",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    }
  }
}
```

Figure 3.6 – Create subnets

3. Launch EC2 Instances

One instance is launched in each subnet.

```
[cloudshell-user@ip-10-6-43-95 ~]$ aws ec2 run-instances --image-id ami-0ef0b498cd3fe129c --count 1 \
> --instance-type t2.micro --subnet-id subnet-0328a7940791d9fc0 \
> --key-name mywestkp --region us-west-2 --no-associate-public-ip-address
{
  "Groups": [],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "ImageId": "ami-0ef0b498cd3fe129c",
      "InstanceId": "i-0288c5d94570afe62",
      "InstanceType": "t2.micro",
      "KeyName": "mywestkp",
      "LaunchTime": "2023-02-14T01:56:33+00:00",
      "Monitoring": {
        "State": "disabled"
      },
      "Placement": {
        "AvailabilityZone": "us-west-2a",
        "GroupName": "",
        "Tenancy": "default"
      },
      "PrivateDnsName": "ip-10-10-2-11.us-west-2.compute.internal",
      "PrivateIpAddress": "10.10.2.11",

```

Figure 3.7 – Create instances

Figure 3.8 shows the resulting VPC architecture.

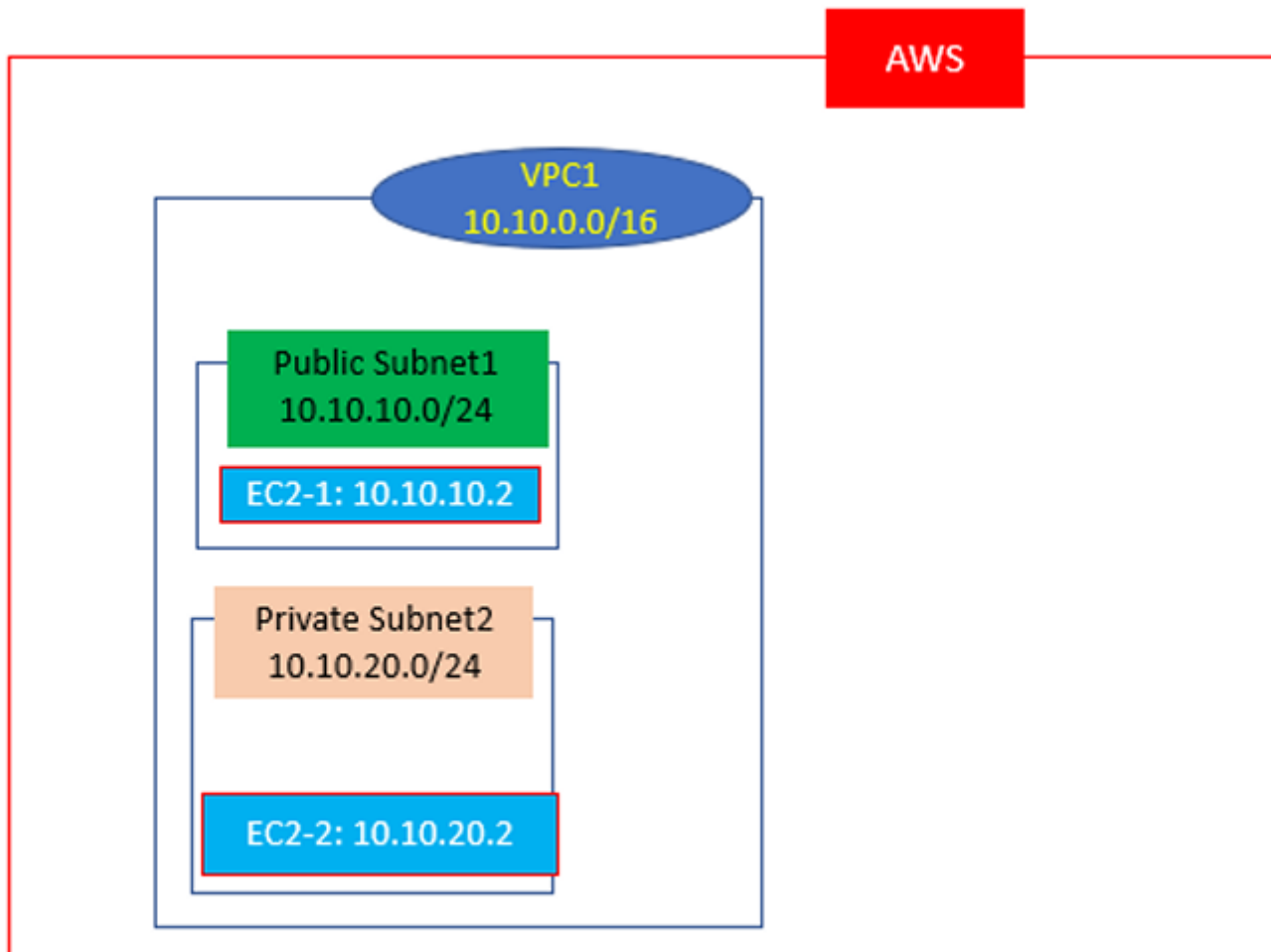


Figure 3.8 – VPC network

Accessing EC2 Instances from the Internet

Public subnet instances require an **Internet Gateway (IGW)** and appropriate routing.

1. Create and Attach an Internet Gateway

VPC > Internet gateways > Create internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IGW1

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q IGW1 X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

Figure 3.9 – Create IGW1

VPC > Internet gateways > Attach to VPC (igw-02afe3e9778325c7e)

Attach to VPC (igw-02afe3e9778325c7e) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0e6043d6091faa7cd X

▶ AWS Command Line Interface command

Cancel

Attach internet gateway

Figure 3.10 – Attach IGW1 to VPC1

2. Update Route Table

Add a default route (0.0.0.0/0) pointing to the IGW.

9 / 22

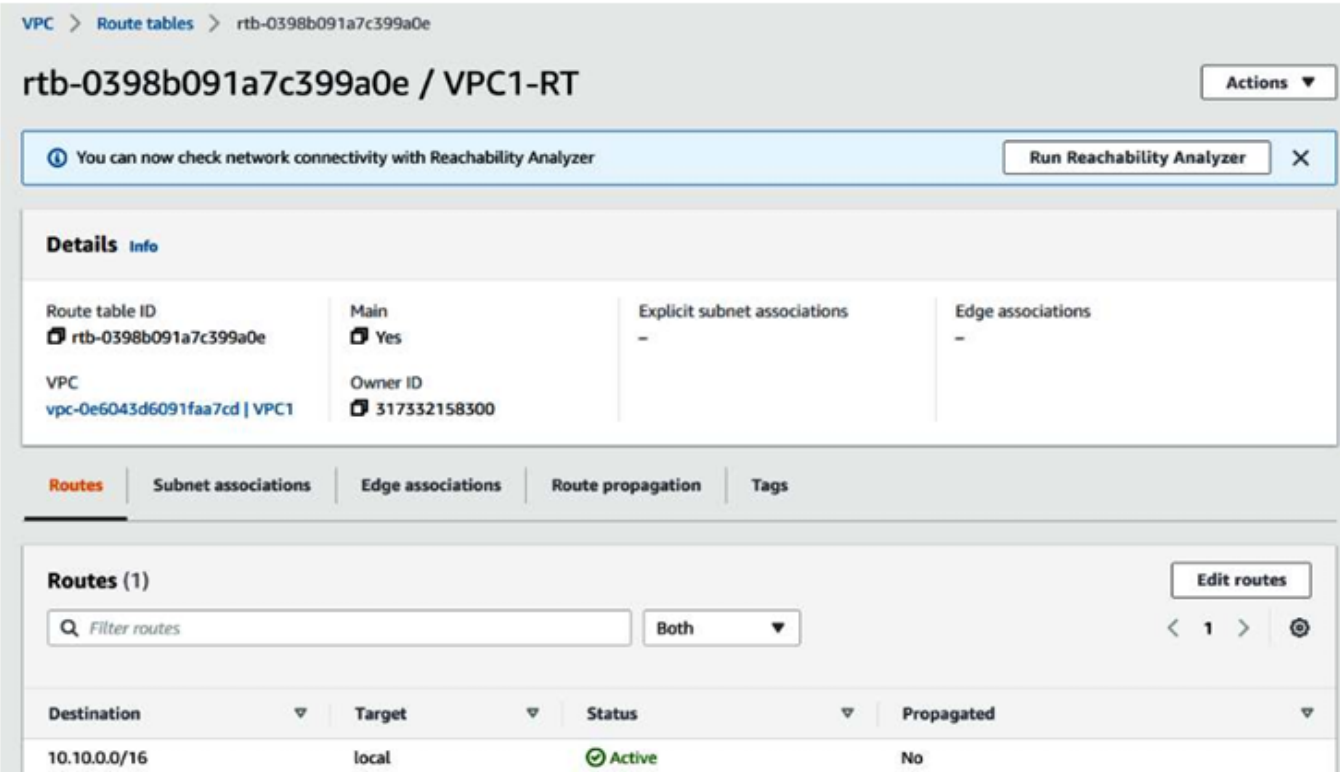


Figure 3.11 – Route table for VPC1

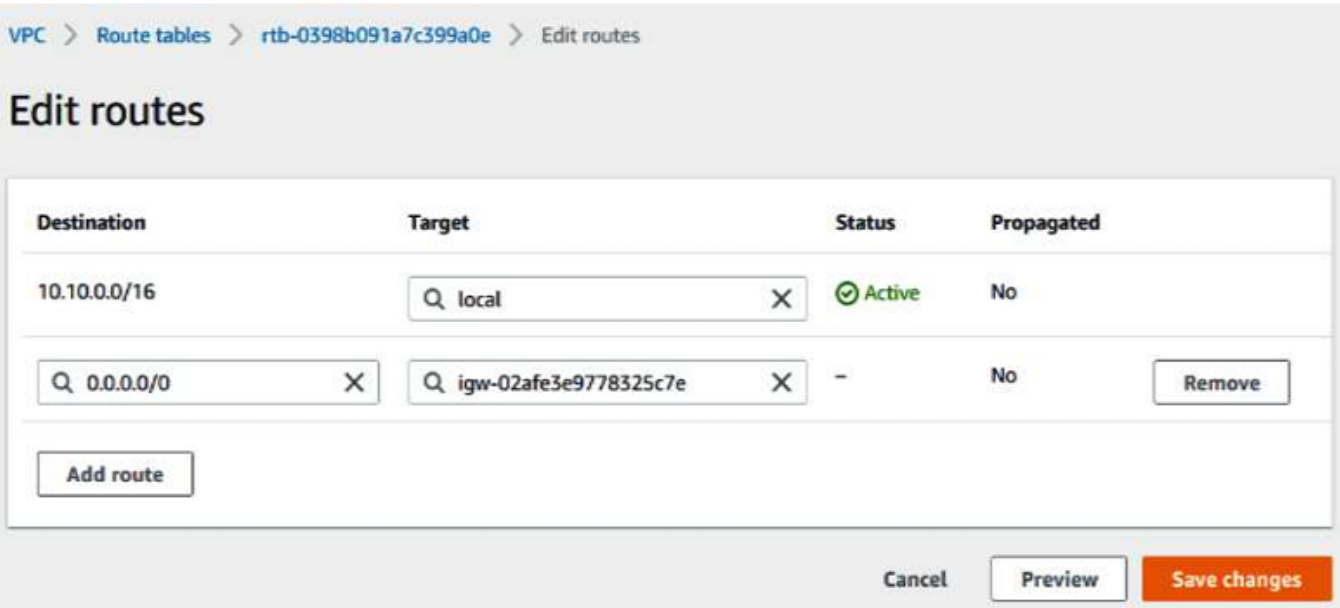


Figure 3.12 – Add internet route to route table

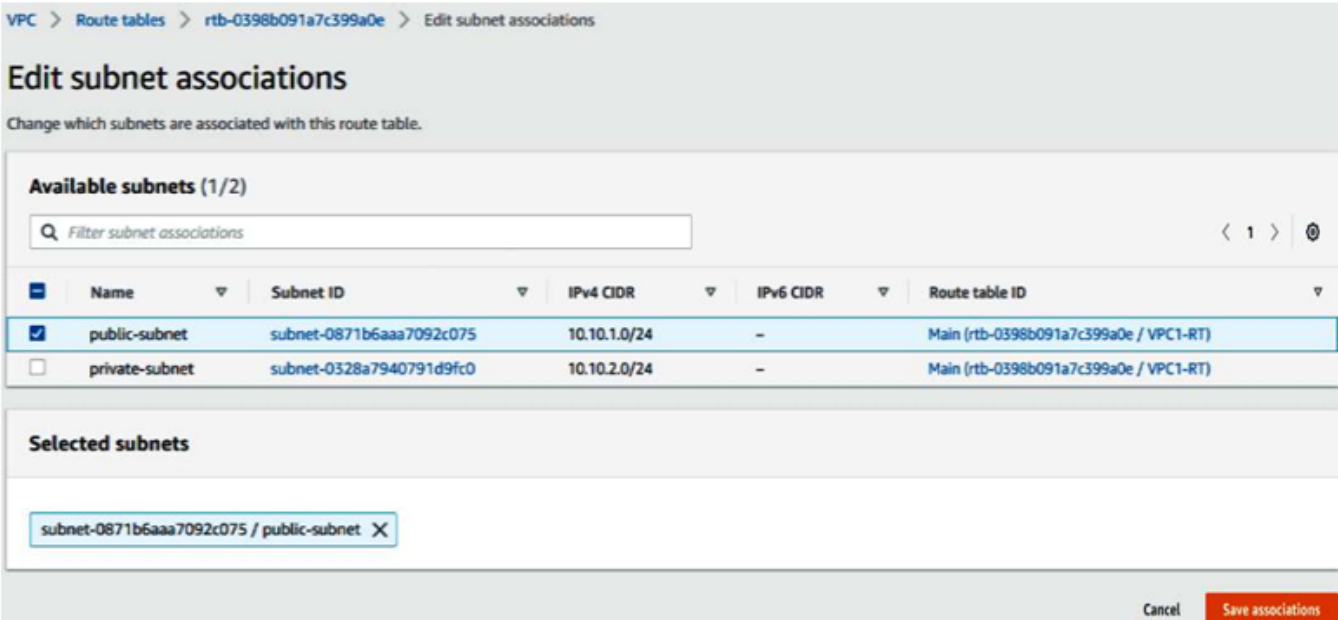


Figure 3.13 – Associate route table to subnet

3. SSH Access

SSH into the public EC2 instance.

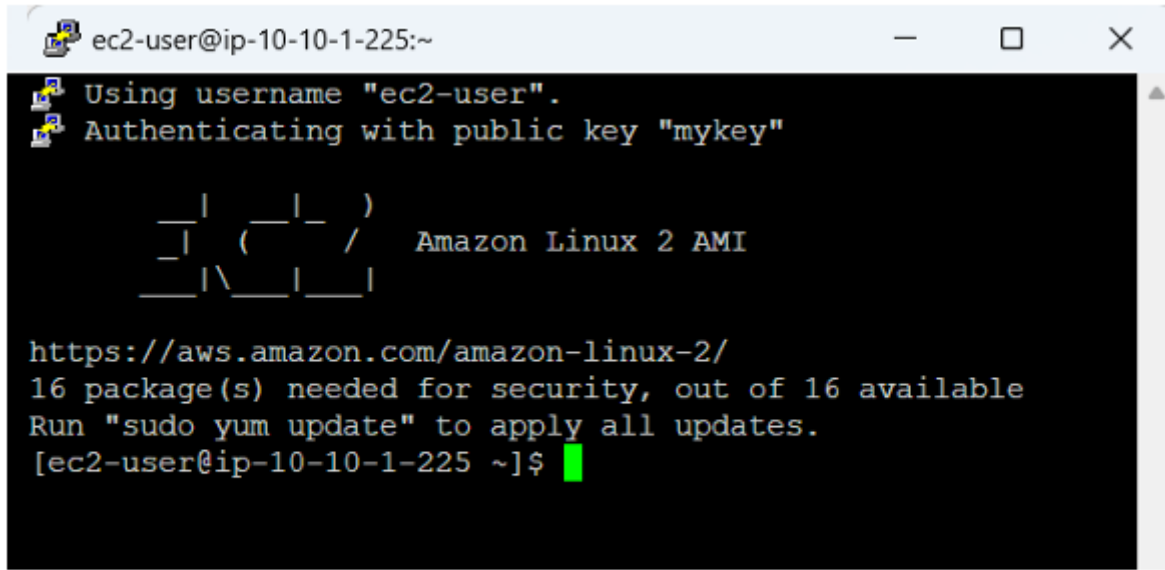


Figure 3.14 – SSH to EC2-1 from the internet

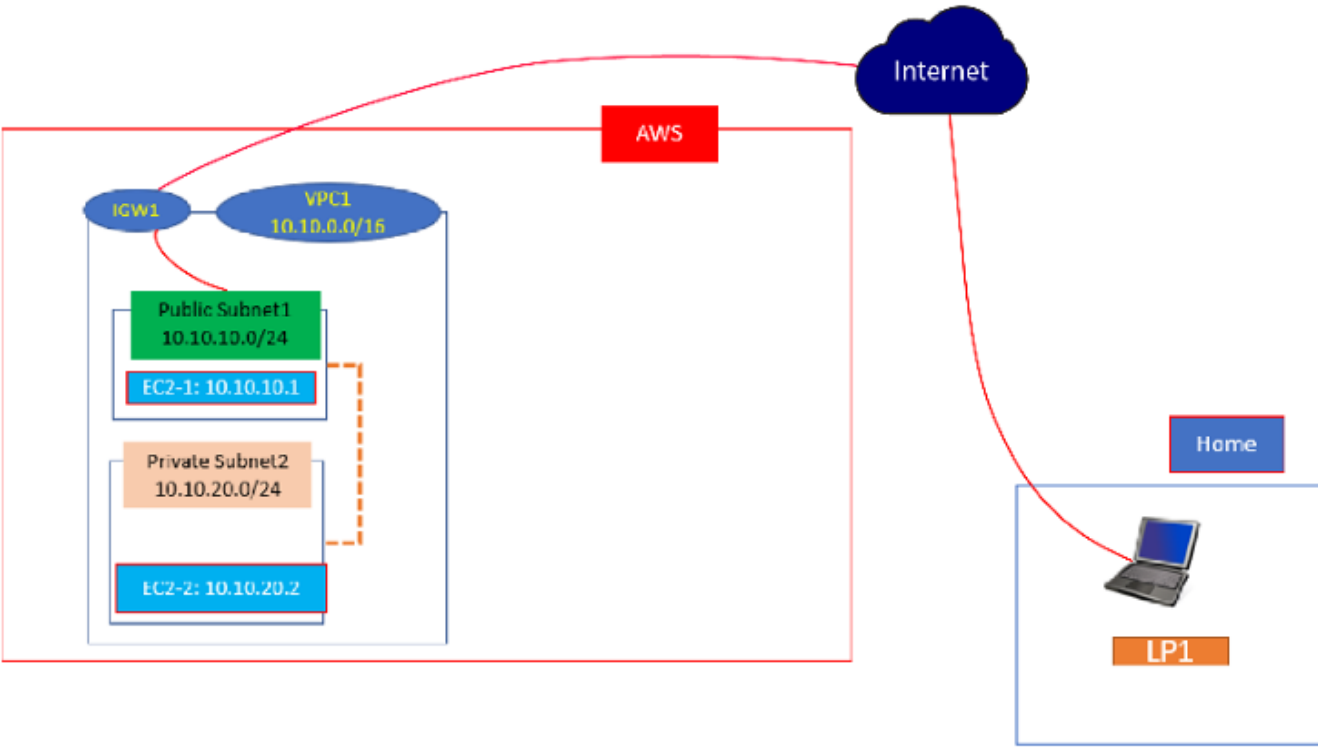


Figure 3.15 – AWS cloud with internet access

Part Two – Connecting Multiple VPCs

Additional VPCs can be connected using **VPC Peering**.

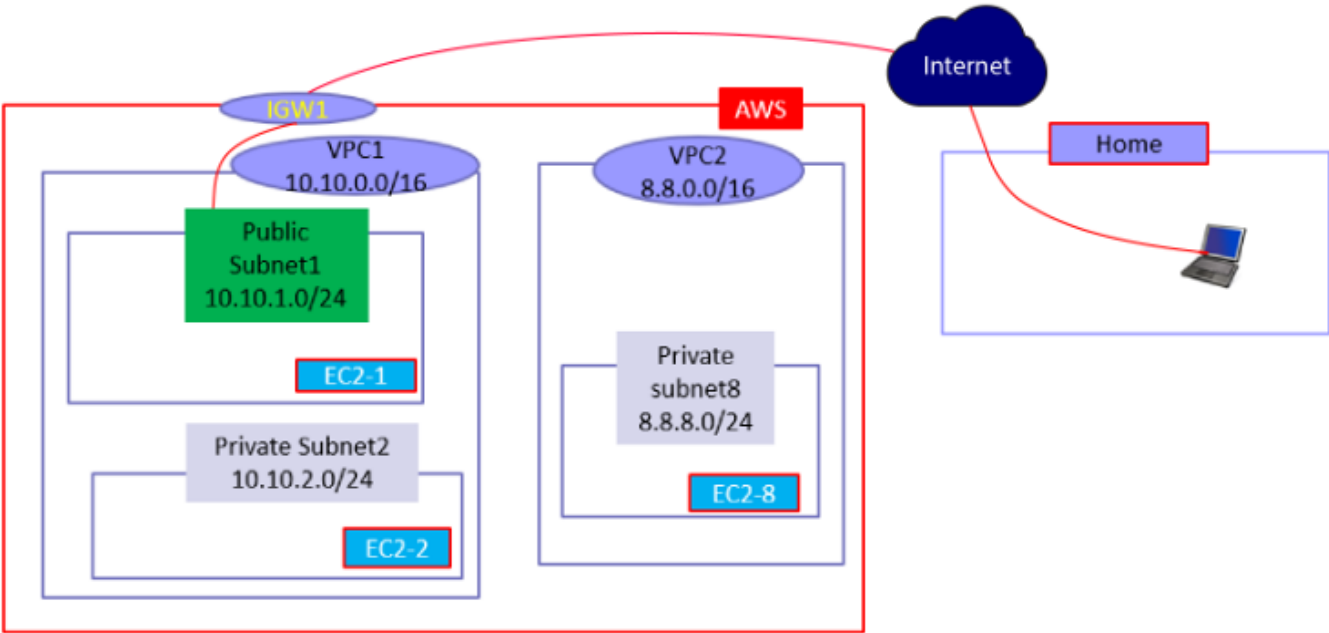


Figure 3.16 – VPC network architecture

Peering the VPCs

Create a peering connection between VPC1 and VPC2.

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

VPC12

Select a local VPC to peer with

VPC ID (Requester)

vpc-0e6043d6091faa7cd (VPC1)

VPC CIDRs for vpc-0e6043d6091faa7cd (VPC1)

CIDR	Status	Status reason
10.10.0.0/16	Associated	-

Select another VPC to peer with

Account

☒ My account

☐ Another account

Region

☒ This Region (us-west-2)

☐ Another Region

VPC ID (Accepter)

vpc-05533af62e30e2168 (VPC2)

VPC CIDRs for vpc-05533af62e30e2168 (VPC2)

CIDR	Status	Status reason
8.8.0.0/16	Associated	-

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q VPC12 X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create peering connection

Figure 3.17 – VPC peering

Update route tables on both VPCs.

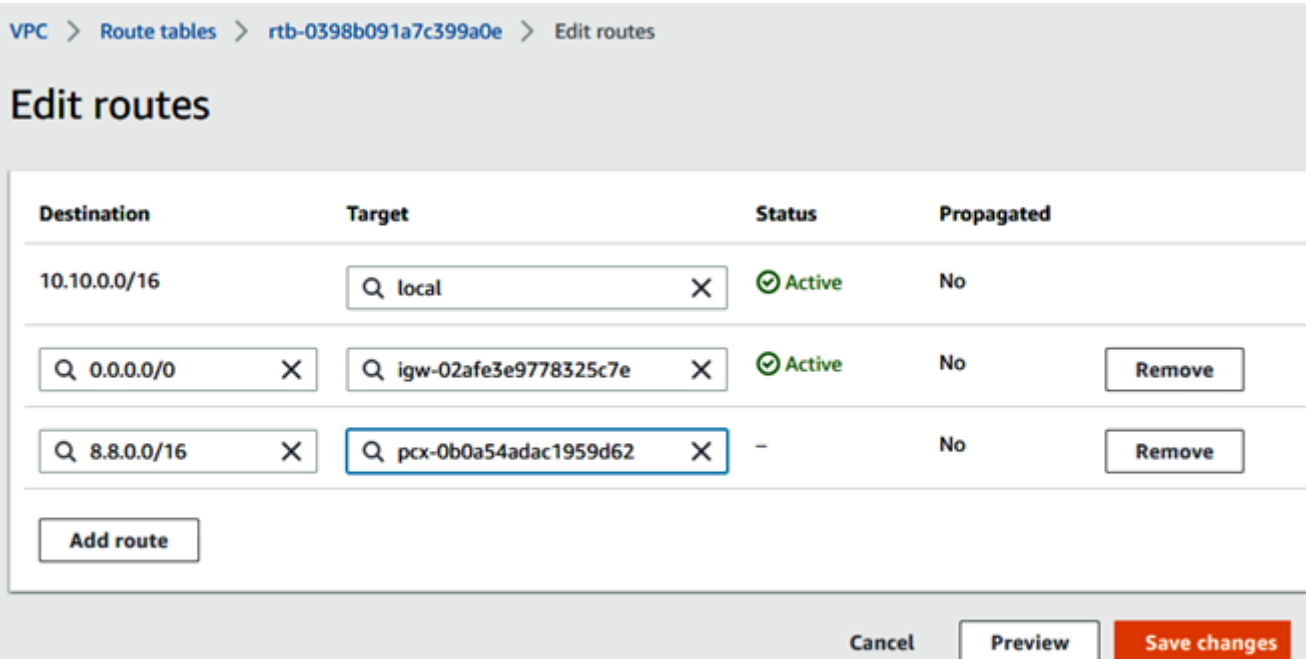


Figure 3.18 – Edit routes for the VPC1 route table

Update security groups to allow ICMP.

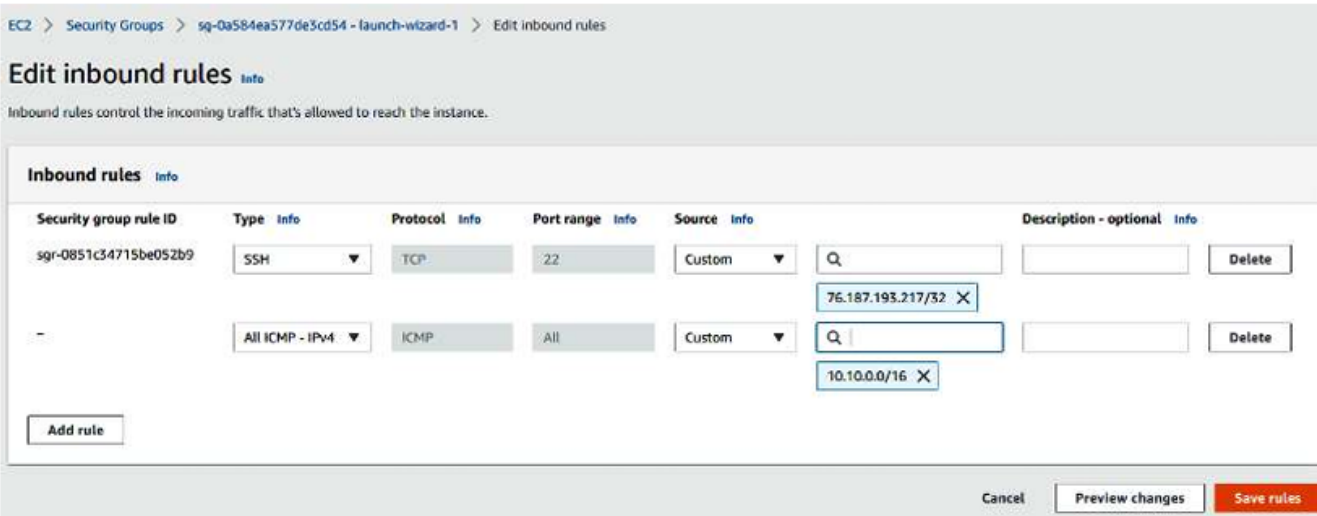


Figure 3.19 – Edit SG inbound rules for EC2-8

```
[ec2-user@ip-10-10-1-225 ~]$ ping 8.8.8.43
PING 8.8.8.43 (8.8.8.43) 56(84) bytes of data.
64 bytes from 8.8.8.43: icmp_seq=1 ttl=255 time=0.955 ms
64 bytes from 8.8.8.43: icmp_seq=2 ttl=255 time=0.553 ms
64 bytes from 8.8.8.43: icmp_seq=3 ttl=255 time=0.495 ms
^C
--- 8.8.8.43 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2023ms
rtt min/avg/max/mdev = 0.495/0.667/0.955/0.206 ms
```

Figure 3.20 – Ping EC2-8 from EC2-1

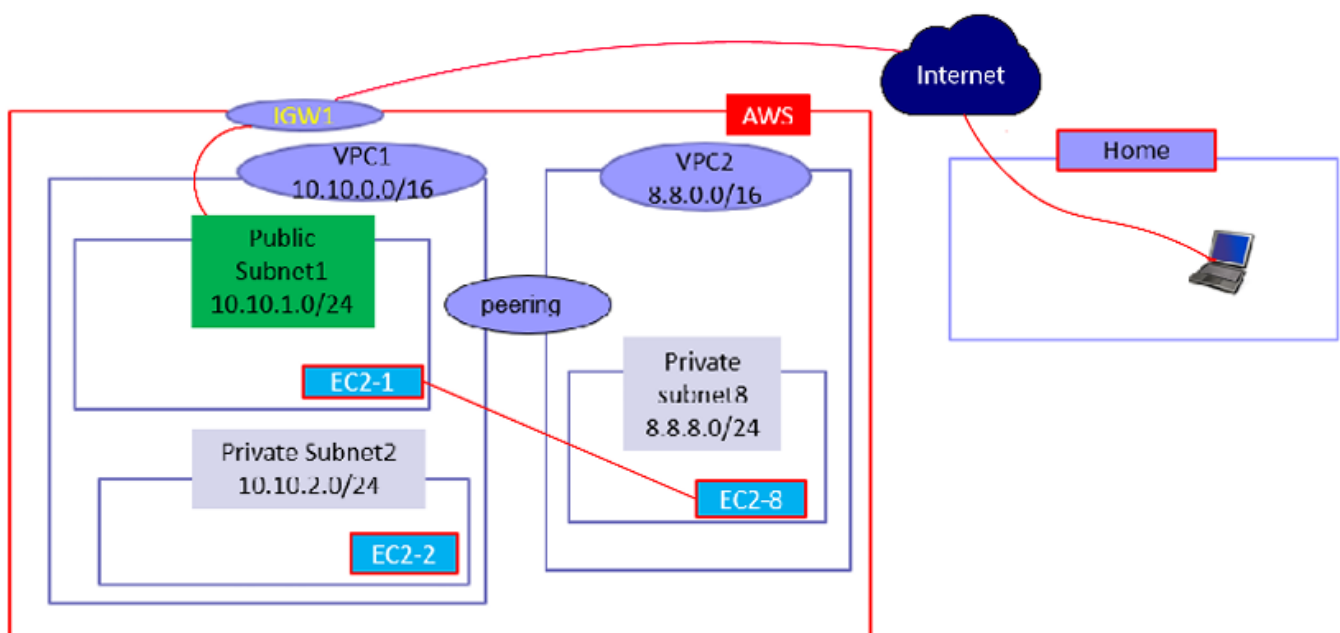


Figure 3.21 – VPC network with peering

Creating a NAT Gateway

A **NAT Gateway** enables outbound internet access for private subnet instances.

Create NAT gateway Info

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the Internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

NAT1

The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-086224ff1a4e9f46f (pn)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.

eipalloc-014fb2aff49fc11a6

Allocate Elastic IP

▶ **Additional settings** Info

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q NAT1 X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create NAT gateway

Figure 3.22 – Create NAT gateway

VPC > Route tables > rtb-097a7f1f134300ab4 > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.10.0.0/16	Q local X	Active	No
Q 0.0.0.0/0 X	Q nat-02f3dd02077a6b028 X	-	No
Add route			

Cancel

Preview

Save changes

Figure 3.23 – Edit routes for the private subnet route table


```
[ec2-user@ip-10-10-1-229 ~]$ ls -l
total 4
-r----- 1 ec2-user ec2-user 1675 Feb 15 16:10 mykey.pem
[ec2-user@ip-10-10-1-229 ~]$ ssh 10.10.2.249 -i mykey.pem
The authenticity of host '10.10.2.249 (10.10.2.249)' can't be established.
ECDSA key fingerprint is SHA256:AYodQL9fBbvlN6XRQLbHSjkNtbsFGevS7CqHSmr2vrk.
ECDSA key fingerprint is MD5:77:e0:ec:a4:40:61:95:68:c3:94:ba:62:98:23:35:7f.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.10.2.249' (ECDSA) to the list of known hosts.

  _ | _ | _ |
  _ | ( _ | _ | / Amazon Linux 2 AMI
  _ | \ _ | _ |

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-10-10-2-249 ~]$ curl www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en"
><head><meta content="Search the world's information, including webpages, images
, videos and more. Google has many special features to help you find exactly wha
```

Figure 3.24 – Access www.google.com from EC2-2

AWS VPC Architecture

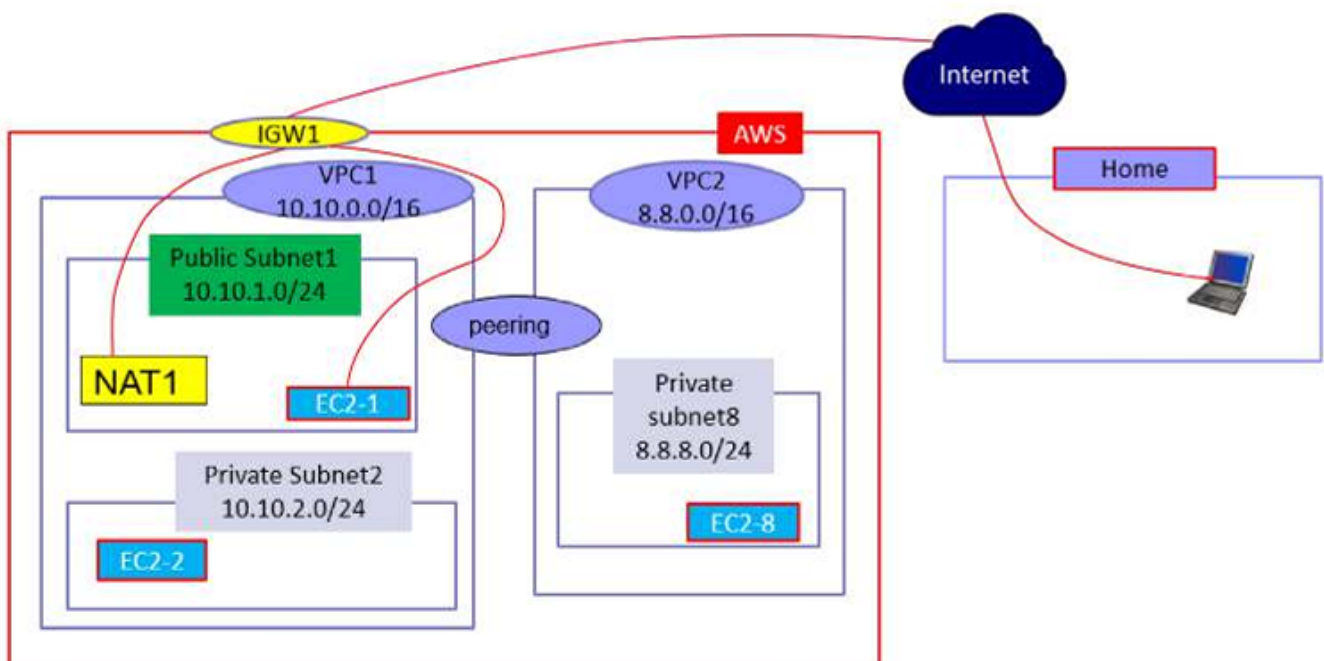


Figure 3.25 – AWS VPC architecture

Part Three – Hardening AWS Network Security

VPC Firewalls

Security Groups (SGs) are stateful, instance-level firewalls.

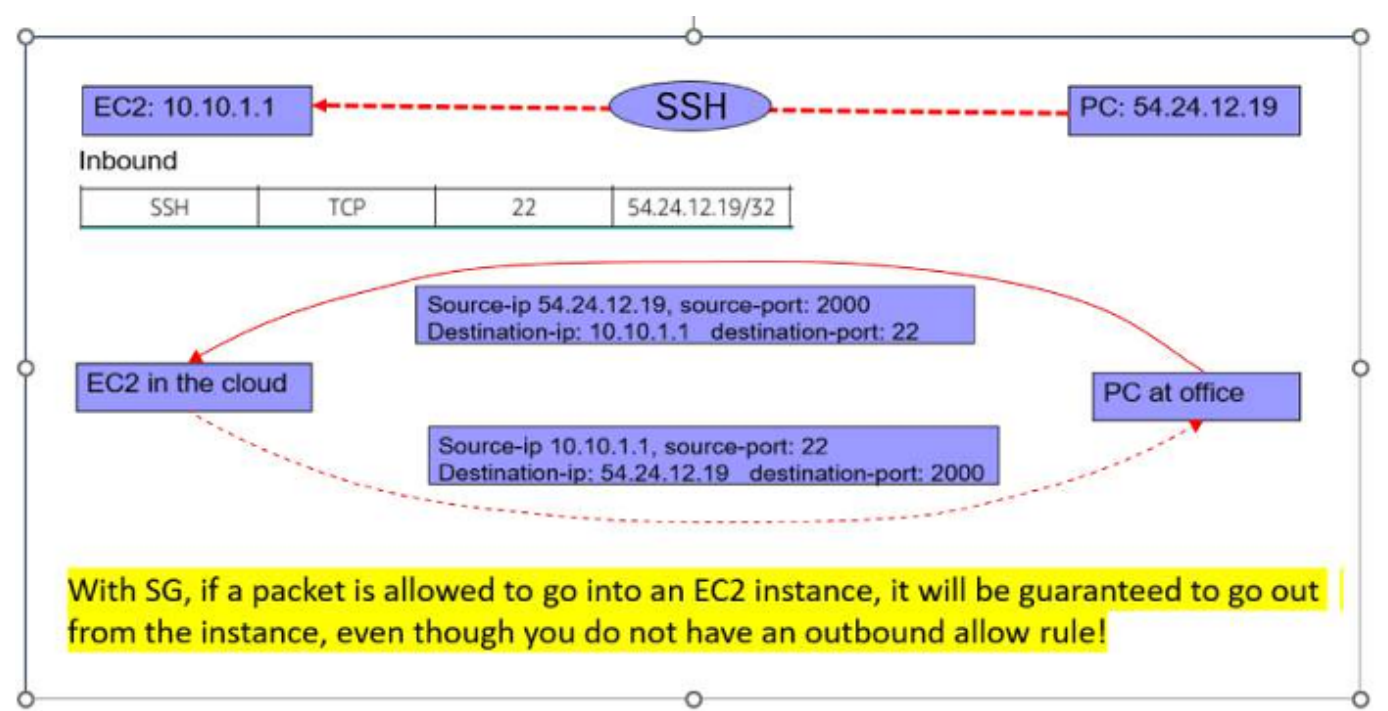


Figure 3.26 – SG is stateful (1)

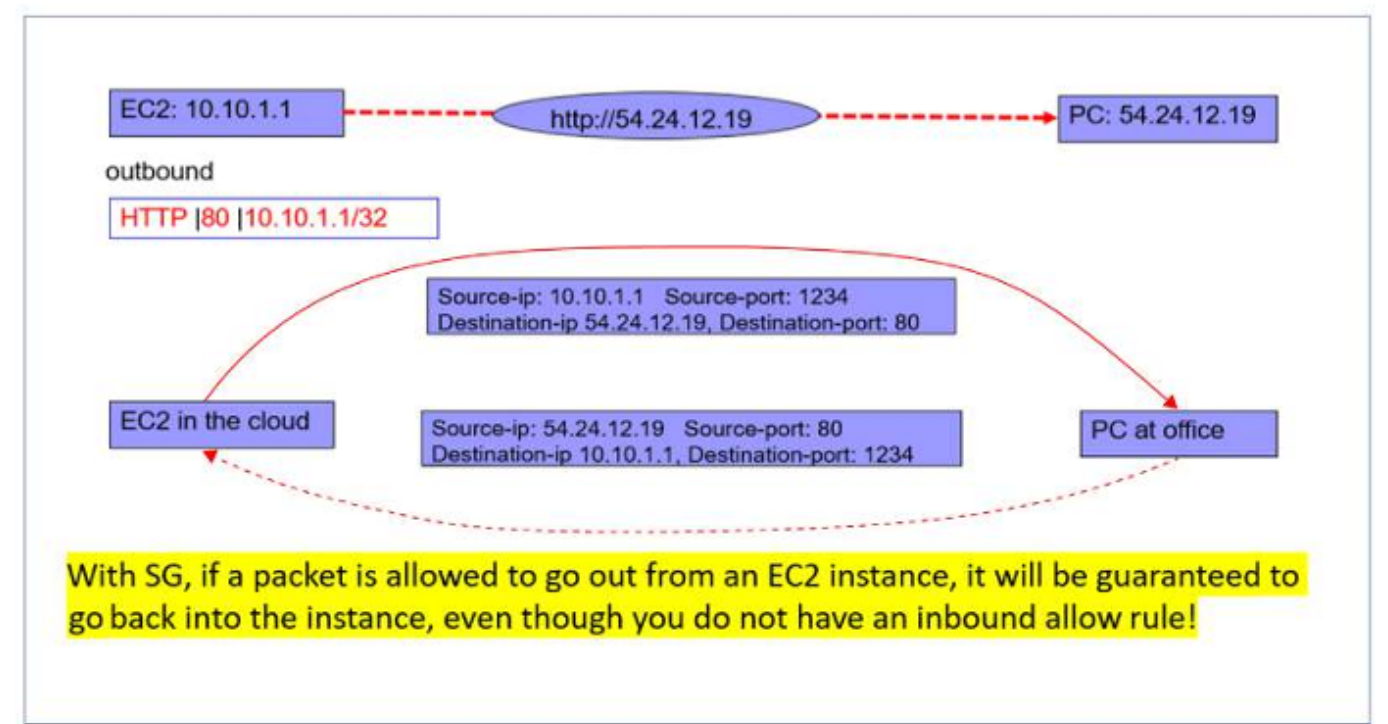


Figure 3.27 – SG is stateful (2)

Network ACLs (NACLs) are stateless, subnet-level firewalls.

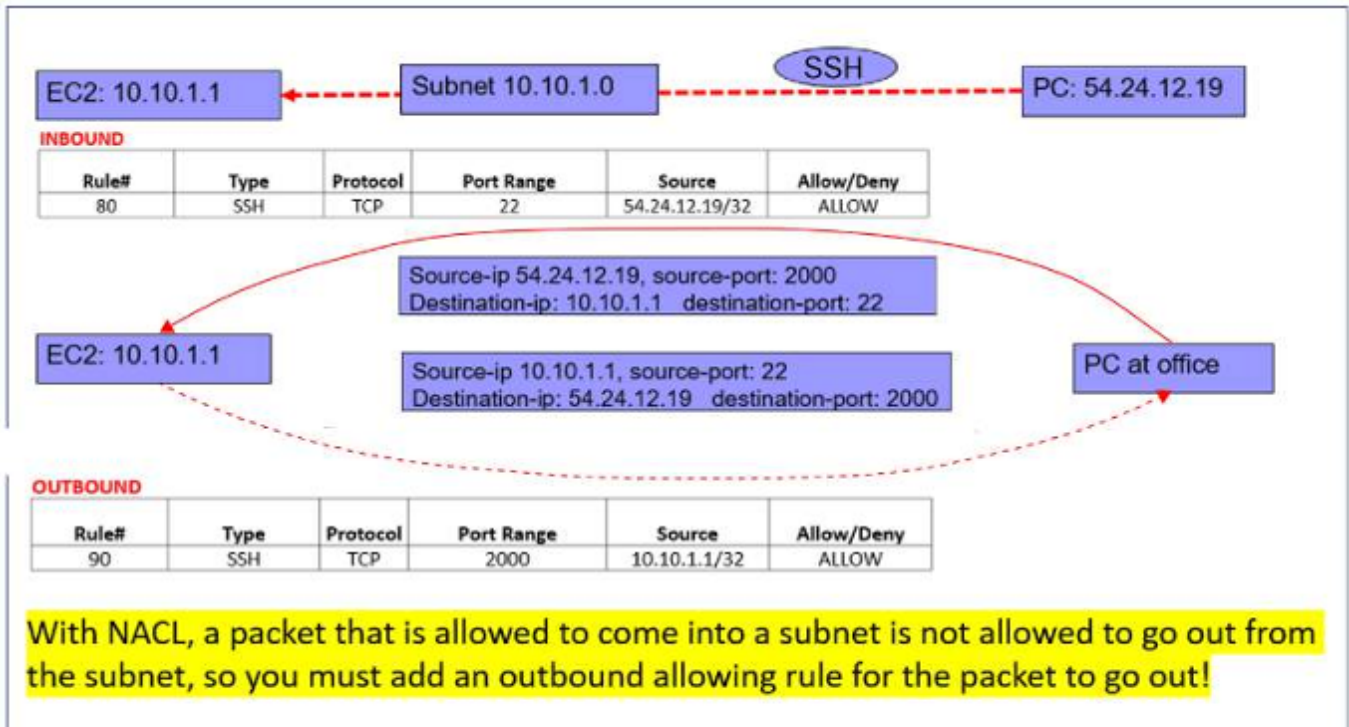


Figure 3.28 – NACL is stateless (1)

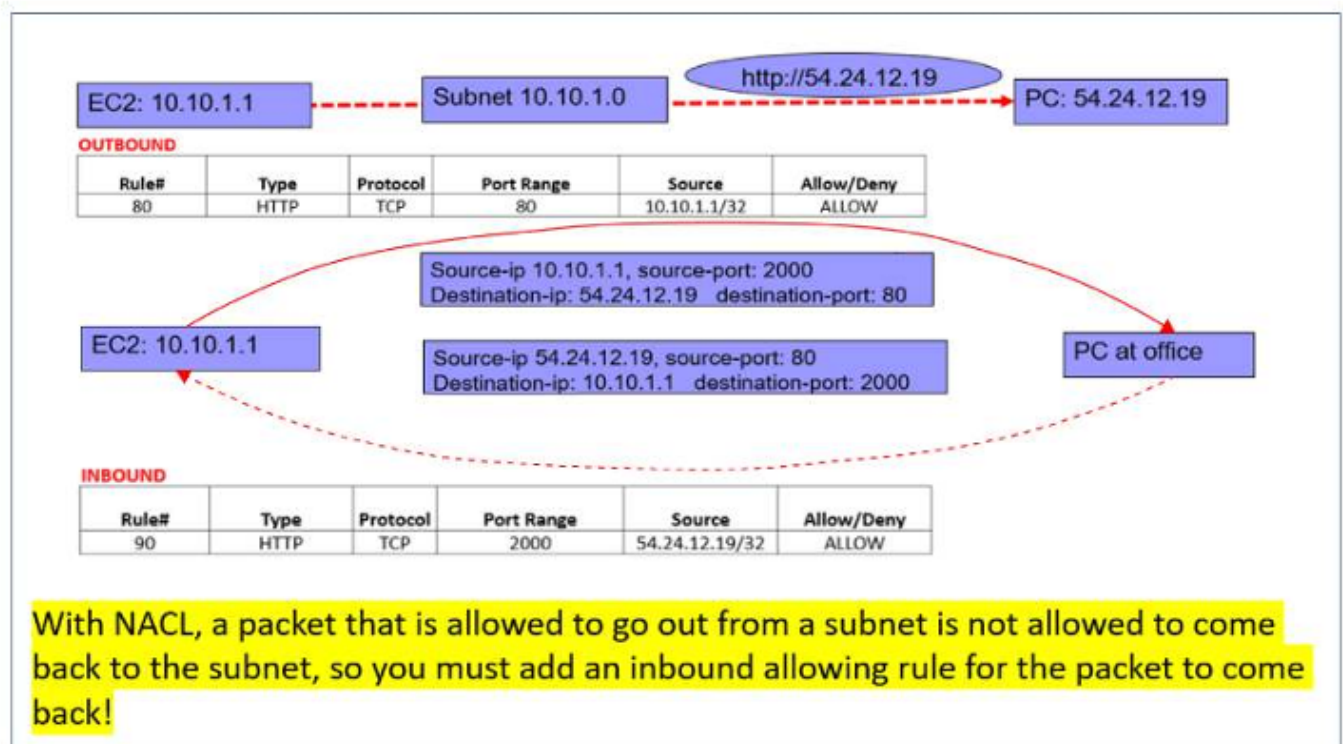


Figure 3.29 – NACL is stateless (2)

VPC Endpoints

VPC endpoints allow private access to AWS services (such as S3) without internet exposure.

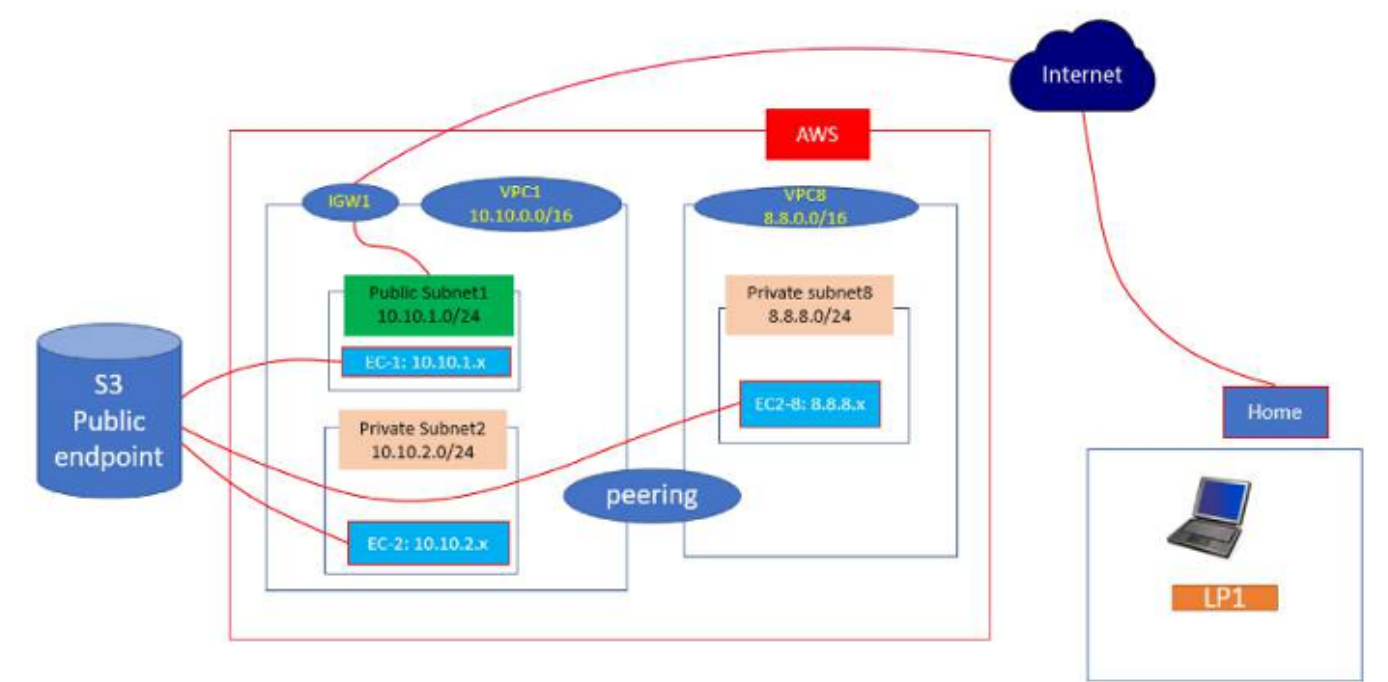


Figure 3.30 – VPC network diagram

Create endpoint info

There are three types of VPC endpoints – Interface endpoints, Gateway Load Balancer endpoints, and Gateway endpoints. Interface endpoints and Gateway Load Balancer endpoints are powered by AWS PrivateLink, and use an Elastic Network Interface (ENI) as an entry point for traffic destined to the service. Interface endpoints are typically accessed using the public or private DNS name associated with the service, while Gateway endpoints and Gateway Load Balancer endpoints serve as a target for a route in your route table for traffic destined for the service.

Endpoint settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-s3-endpoint

Service category
Select the service category.

☒ AWS services
Services provided by Amazon

☐ PrivateLink Ready partner services
Services with an AWS Service Ready designation

☐ AWS Marketplace services
Services that you've purchased through AWS Marketplace

☐ Other endpoint services
Find services shared with you by service name

Services (1/3)

Filter services

Service Name: com.amazonaws.us-east-1.s3 Clear filters

Service Name	Owner	Type
☒ com.amazonaws.us-east-1.s3	amazon	Gateway
☐ com.amazonaws.us-east-1.s3	amazon	Interface
☐ com.amazonaws.us-east-1.s3-outposts	amazon	Interface

VPC

Select the VPC in which to create the endpoint.

VPC

The VPC in which to create your endpoint.

vpc-0d90cde16d2e815a2 (VPC1)

Route tables (1/2) info

Filter route tables

Name	Route Table ID	Main
☐ VPC1RT	rtb-0595676d643322bb5 (VPC1RT)	Yes
☒ VPC1-Pr	rtb-0fa2616ffb9955b5c (VPC1-Pr)	No

Figure 3.31 – VPC endpoint settings

20 / 22

Understanding Amazon Direct Connect

AWS Direct Connect provides a private, dedicated connection between on-premises data centers and AWS with predictable performance.

Setup steps include requesting a connection, selecting a partner, configuring routers, and validating connectivity.

Understanding Amazon DNS – Route 53

Amazon Route 53 is a scalable DNS service offering:

- Domain name resolution
- Traffic routing
- Health checks and failover

Supported routing policies include simple, weighted, latency-based, geolocation, failover, and multivalue routing.

Understanding the Amazon CDN

Amazon CloudFront is AWS’s global CDN service, providing secure, low-latency delivery of static and dynamic content through worldwide edge locations.

Essential Commands Explained

Command	Purpose
<code>aws ec2 create-vpc</code>	Creates a new VPC with a specified CIDR block
<code>aws ec2 create-subnet</code>	Creates a subnet within a VPC
<code>aws ec2 run-instances</code>	Launches EC2 instances
<code>aws ec2 create-internet-gateway</code>	Creates an Internet Gateway
<code>aws ec2 create-nat-gateway</code>	Enables outbound internet access for private subnets
<code>ping</code>	Tests network connectivity
<code>ssh</code>	Securely connects to EC2 instances
<code>curl</code>	Tests outbound internet access

Reference Links

- AWS VPC: <https://aws.amazon.com/vpc/>
- AWS EC2: <https://aws.amazon.com/ec2/>

- AWS Direct Connect: https://docs.aws.amazon.com/directconnect/latest/UserGuide/getting_started.html
 - VPN over Direct Connect: <https://aws.amazon.com/premiumsupport/knowledge-center/create-vpn-direct-connect/>
 - Amazon Route 53: <https://aws.amazon.com/route53/>
 - Amazon CloudFront: <https://aws.amazon.com/cloudfront/>
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